Filters

Intro to Audio Tech 1
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Lesson Outline

- Frequency Response
- Impulse Response
- Max Demo
- Up Next…

Impulse Response

- We can send an impulse through our filter to determine how it reacts in the **time domain**
- A perfect impulse is a short burst signal containing all frequencies within a sampling rate
  - This looks like a signal of [1, 0, 0, 0, 0, 0…]
- For LTIs, the impulse response fully characterizes the filter
- For the "Simplest Possible Filter":

$$y[n] = \mathrm{x[n]} + x[n-1]$$
$$\mathrm{h}[n] = \{1, 1, 0, 0, 0, 0..\}$$

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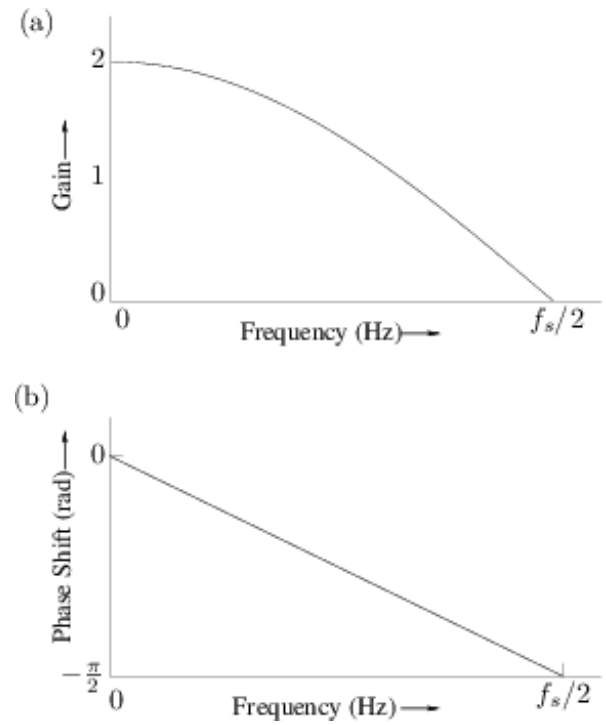
Frequency Response

- We can send the same unit impulse through a filter to determine how it acts in the **frequency domain**
- However, this time instead of looking at how the input changes over time, we are looking at how the spectrum changes
- The unit impulse contains a flat spectrum of all frequencies, so we find the frequency response to determine how those frequencies are attenuated
- You'll see this in two graphs: magnitude frequency response (a) and phase response (b)



Impulse vs Frequency Response

- Impulse Response
  - The filter's response to a unit impulse – how it characterizes that impulse
  - "How are the outputted samples affected"
  - Time domain
- Frequency Response
  - The filter's response to a unit impulse
  - How does the system respond to specific frequencies
  - Frequency domain

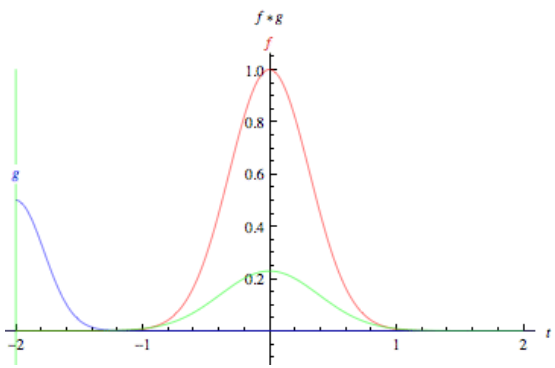
Finite Impulse Response vs Infinite Impulse Response

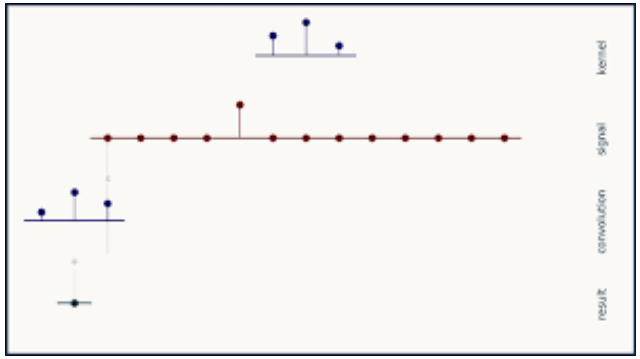
- FIR
  - Simple delay line
  - Filter's response to an impulse dies over a **finite** period of time
  - Pros: "Easy" design, stable, linear phase
  - Cons: Not efficient, more memory used and longer delay, requires high order for "good" filtering
  - Example FIR High pass Filter:
    - $y[n] = x[n] - x[n-1]$
- IIR
  - Feedback, recursive
  - Filter's response feeds back on itself
  - Pros: efficient, lower order for sharp filtering, quick response
  - Cons: Unstable (can "blow up"), phase distortion and nonlinear phase
  - Example IIR High pass Filter:
    - $y[n] = x[n] - y[n-1]$

Convolution

Convolution

- Expresses how the “shape” of one function modifies the other
  - An integral that expresses the amount of overlap of one function as its shifted over another
  - ”Blends” two functions
- $(f * g)(t)$





Convolution

- Calculate f*g, the convolution of the signals f=[1 2 -1 3 2] and g=[3 4 2].

1	2	-1	3	2

*

3	4	2

Convolution

1	2	-1	3	2

*

3	4	2

1	2	-1	3	2

2	4	3

Convolution

1	2	-1	3	2

*

3	4	2

1	2	-1	3	2

2	4	3

1(3) = 3

3						

Convolution

1	2	-1	3	2

*

3	4	2

	1	2	-1	3	2
2	4	3			

1(4)+2(3) = 10

3	10					

Convolution

1	2	-1	3	2

*

3	4	2

1	2	-1	3	2

2	4	3

1(2)+2(4)+(-1)(3) = 7

3	10	7				

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Convolution

1	2	-1	3	2

*

3	4	2

1	2	-1	3	2
	2	4	3	

2(2)+(-1)(4)+3(3) = 9

3	10	7	9			

Convolution

1	2	-1	3	2

*

3	4	2

1	2	-1	3	2
		2	4	3

-1(2)+3(4)+2(3) = 16

3	10	7	9	16		

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Convolution

1	2	-1	3	2

*

3	4	2

1	2	-1	3	2

2	4	3

3(2)+2(4) = 14

3	10	7	9	16	14	

Convolution

1	2	-1	3	2

*

3	4	2

1	2	-1	3	2

2	4	3

2(2) = 4

3	10	7	9	16	14	4

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